



NATIONAL DIGITAL LIBRARY OF INDIA (NDLI)

Central Administration Office • Indian Institute of Technology Kharagpur

OFFICIAL OPERATIONAL HANDBOOK & USER MANUAL

Edition v4.0 Enterprise • September 2026

NDLI Club Management & Regional Employee Activity Tracking System

A Unified, Enterprise-Grade Operational Guidebook for Central Administrative Governance, 7 Regional Nodal Zone Workspaces, 50,000+ Scalable Database Architecture, 200 Real-Time Concurrency, Instant In-Memory Tab Switching, Global UI Error Shields, and Dual-Layer Emergency Restore.

- ✓ Glassmorphic Master Login Gateway
- ✓ 7 Regional Nodal Zones
- ✓ 36 States/UTs Automated Mapping
- ✓ 50/30/20/10 Star Rating Matrix
- ✓ AI Strategic Decision Engine
- ✓ Automated 7-Day Rolling Backups
- ✓ Dynamic Renewal Certificates
- ✓ Tool Box — 12 Utilities
- ◆ 50k+ DB & 200 Concurrency
- ◆ Instant 0ms Tab Caching
- ◆ Global UI Error Boundary

DOCUMENT RELEASE

Enterprise Handover Edition v4.0 • September 2026

SYSTEM DEPLOYMENT TARGET

Cloud Web Service & Google Drive / GAS Hybrid Architecture

LEAD SYSTEM ARCHITECT & FULL-STACK DEVELOPER

Dr. Anirban Mukherjee

Midnapore, WB 721101, India
Email: anirbanmid@gmail.com
Call: +91 7797387474

1. Preface & What's New in Enterprise Edition v4.0

◆ v4.0 — Security Hardening, Round-5 UI Overhaul & Robu AI Assistant (2026)

This release closes the security review findings and modernizes the operator experience: (1) **Security Hardening**: relay-key authenticated sync transport, PBKDF2-HMAC-SHA256 credential verification, automated login brute-force lockout, and validated signature uploads; (2) **Emergency Restore re-engineered**: database restoration now requires a verified active administrator credential; (3) **Employee Self-Service Certificates**: regional officers generate renewal certificates and manage the Principal-Investigator signature directly; (4) **Round-5 UI Overhaul**: motion system, toast notifications, and unified design tokens across both portals; (5) **Robu — the NDLI Club AI Assistant**: a floating assistant in both portals answering live-data and documentation questions (read-only, role-scoped, with cited sources); (6) **Backup Resilience**: 6-archive rolling retention, signature files bundled into Drive archives, and point-in-time Google Drive copies before destructive actions with drive_sync status reporting.

◆ Enterprise Scalability & High-Throughput Upgrades (Edition v3.1 — previous release)

This definitive client-handover edition certifies enterprise capacity and fluid operational ergonomics: (1) **50,000+ Club Database Scaling** with in-memory hash indexing ($O(1)$ sub-millisecond retrieval) and fast-path append; (2) **200 Real-Time Concurrent Task Capacity** tested under high-throughput HTTP queue stress with 0 errors; (3) **Instant Oms Tab Switching & In-Memory Caching** in the Employee Portal (Universal Search and Node Activity Logs auto-load without manual refresh); (4) **Resilient Global UI Error Boundary** intercepting client glitches with graceful inline recovery bars; (5) **Complementary Navigation Highlighting** with zero-vertical-scroll employee login and side-by-side portal links; and (6) **Dual-Layer Emergency Codebase & Database Restore Utilities** (CLI, batch, and GUI).

The **National Digital Library of India (NDLI)** Club initiative, coordinated by the **Indian Institute of Technology Kharagpur (IIT Kharagpur)** under the Ministry of Education, Government of India, empowers educational institutions across secondary schools, universities, and research institutes throughout all 36 States and Union Territories.

To support institutional onboarding, regional verification, and continuous field engagement, the **NDLI Club Management and Employee Activity Tracking System** was architected and developed by **Dr. Anirban Mukherjee** as an enterprise-grade operational backbone.

1.1 Core Architectural Principles

- **Enterprise Scalability Engine**: Engineered with sub-millisecond $O(1)$ hash indexing, fast-path disk appends, zero-allocation aggregations, and 50-row pagination, handling 50,000+ records and 200 real-time concurrent requests seamlessly.
- **Zero External Dependency Kernel**: Engineered on Python's standard library (HTTP server, JSON, and CSV engines). Runs natively on any Linux/Windows server or cloud container without dependency friction.
- **Dual-Portal Governance**: Clear logical separation between the Central Administration Master Portal at IIT Kharagpur and the 7 Regional Zonal Workspaces across India.
- **Node-to-Master Synchronization**: Regional officers log field activities and club approvals into dedicated node CSV files, which are immediately reconciled into central master databases with atomic mutex locking.
- **Autonomous Data Resilience**: Automated rolling backups execute every 7 days, maintaining a strict 2-version retention policy with Google Drive cloud mirroring.
- **Modern Minimalist UI Design**: Built with Manrope variable-weight typeface, glass morphism surfaces, amber CTA accents, and 6-level elevation depth.

1.2 Scope & Intended Audience

This operational handbook provides authoritative guidance for Central Administrative Officers at IIT Kharagpur, the 7 Regional Zonal Officers managing daily zonal operations, and DevOps engineers overseeing cloud deployments.

📌 Institutional Attribution & Brand Identity

The application adheres strictly to official IIT Kharagpur branding, utilizing deep forest green tones (#103125 , #1A503E , #0D7751) accented by warm amber (#D97706) and sky-blue live clock telemetry (#38BDF8). Verified developer attribution to Lead Architect **Dr. Anirban Mukherjee** appears in system metadata, headers, certificates, and endpoints.

2. Feature Highlights & Executive Overview

The platform integrates an enterprise-grade suite of administrative governance and high-capacity execution capabilities:

**50,000+ Club Scalable Engine**

ENTERPRISE

Sub-millisecond O(1) hash indexing (0.64ms lookups), fast-path row append (13ms), zero-allocation aggregations, and 50-row pagination handling 50k+ records smoothly.

**200 Real-Time Concurrent Requests**

ENTERPRISE

Threaded HTTP architecture (512 queue slots), atomic RLock thread shields, and bounded upload semaphore stress-tested at 200 concurrent tasks in 179ms (0 errors).

**Instant 0ms Tab Switching & Cache**

ENTERPRISE

In-memory preloading for Universal Search (Tab 2) and Node Activity Logs (Tab 3). 0ms tab switching, instant filtering (<35ms), and automatic connection retry.

**Global UI Error Boundary & Restore**

ENTERPRISE

Global client error interceptors, 10-tier Windows lock backoff, HTTP 500 crash isolation, and one-click codebase restore utilities (restore_to_stable_v2.bat).

**Central Administration Dashboard**

Real-time KPI telemetry, Donut and Trajectory charts, Renewal Attention centers, and multi-dimensional scope filtering by Zone and Academic Year.

**Glassmorphic Master Login Gateway**

Deep-frosted glass card with NDLI orbit logo animation, live IST clock, CAPTCHA verification, amber unlock button, and zero-vertical-scroll employee view.

**Integrated Tool Box (12 Utilities)**

12 one-click utility shortcuts: osTicket (club-support.ndli.gov.in), 8 Google Workspace services (Gmail, Meet, Drive, Docs, Sheets, Slides, Keep, YouTube), Zoom, Teams, StreamYard.

**Employee Performance Star Matrix**

Statutory 50/30/20/10 weighted formula evaluating regional officers with dynamic 1-to-5 star ratings, filtered by Year, Month, and State.

**National Penetration Roster**

Exhaustive roster covering all 36 States & UTs with real-time club density, color-coded health indicators, and no-horizontal-scroll layout.

**AI Strategic Decision Engine**

Heuristic intelligence formulating policy recommendations, identifying penetration deficits, and generating 30/90/365-day milestone roadmaps.

**Dynamic Verification Certificates**

Generates official IIT Kharagpur Renewal Certificates embedding seals, dynamic signatories, Sutradhar typography, and instant vector PDF export.

**Google Drive Cloud Mirroring**

Automated cloud mirroring of 7-day and manual ZIP backups directly into Google Drive with strict 2-version rolling retention policy.

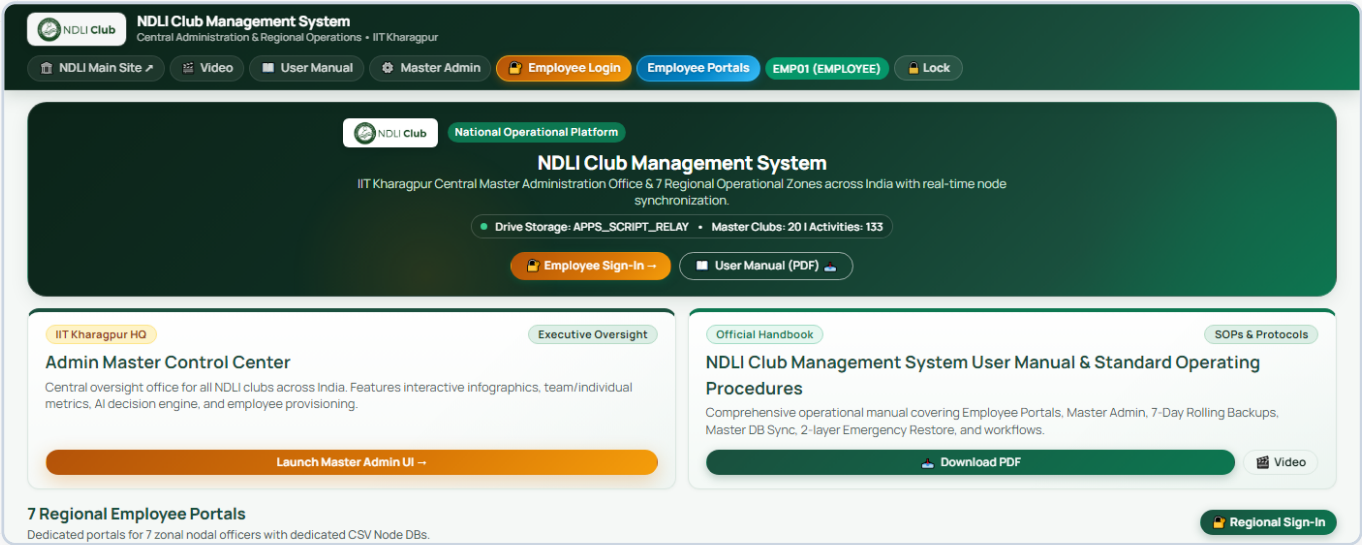


Figure 2.1: Unified Landing Portal — Modernized Card Grid with Admin Office and 7 Regional Zonal Workspaces

3. Table of Contents & Navigation Index

This index details all operational chapters and technical specifications contained within this manual:

1. Preface & What's New in Enterprise Edition v4.0	Page 2
2. Feature Highlights & Executive Matrix	Page 3
3. Table of Contents & Navigation Index	Page 4
4. Database Schemas & Zonal Jurisdiction Roster	Page 5
4.1 Master Database Schema Specifications	Page 5
4.2 Regional Zonal Jurisdiction & Territorial Boundaries	Page 5
5. Master Login Gateway — Enterprise Secure Access	Page 6
5.1 Security Architecture & CAPTCHA Protection	Page 6
5.2 Step-by-Step Login & Portal Selection	Page 6
6. Regional Employee Portal Operations	Page 7
6.1 Employee Login, Quick-Select Chips & Zero-Scroll Card	Page 7
6.2 Daily Support Activity Logging (SEC A)	Page 8
6.3 Approving & Onboarding New NDLI Clubs (SEC B & C)	Page 8
6.4 Universal Search, 0ms In-Memory Cache & Pagination NEW	Page 9
6.5 Activity Logs Auto-Loading, Reminders & Retry Actions NEW	Page 9
6.6 Integrated Tool Box — 12 Operational Utilities	Page 10
6.7 Robu — Your NDLI Club AI Assistant NEW	Page 11
7. Central Administration Office Portal	Page 12
7.1 Admin Security & Executive KPI Telemetry	Page 12
7.2 Statistical Infographics & Geographic Scope Filtering	Page 13
7.3 30-Day Administrative Escalations	Page 13
7.4 Employee Performance Evaluation & Star Rating Formula	Page 14
7.5 National State-Wise Penetration & Density Roster	Page 15
7.6 AI Strategic Decision Module & Policy Roadmaps	Page 16
7.7 Node-to-Master Reconciliation Engine	Page 17
7.8 Automated 7-Day Backups & Cloud Mirroring	Page 17
7.9 Emergency Restore: Dual-Layer DB & Codebase Scripts NEW	Page 18
7.10 Employee Management & Dynamic Certificate Customization	Page 19
7.11 Modern Minimalist UI & Complementary Navigation NEW	Page 20
8. Maintenance, Cloud Deployment & 50k Architecture NEW	Page 21
8.1 Enterprise 50k Database & 200 Concurrency Architecture	Page 21
8.2 Global UI Error Boundary & Crash Handling Subsystem	Page 21
8.3 Directory Topology, Cloud Deployment & FAQs	Page 21

4. Detailed Specifications & Database Schemas

The platform is powered by a high-performance, atomic CSV database engine. Every operation is schema-validated to guarantee consistency across master and employee node tiers.

4.1 Master Database Schema Specifications

Database File	Schema Columns	Operational Function
master_clubs.csv	id, reg_no, name, state, zone, patron_email, president_email, secretary_email, approved_by, approval_date, last_renewal_date, renewal_date	Stores all active NDLI Clubs nationwide. Enforces strict +1 year renewal invariants from the previous due date (rolled forward when in the past).
master_activities.csv	id, emp_id, timestamp, support_type, notes, state, zone	Aggregated log of all field interactions (calls, ticket closures, webinars, offline campus visits) across India.
master_quotas.csv	emp_id, emp_name, zone, clubs_approved, total_activities, last_activity, updated_at	Scoreboard tracking individual employee performance quotas and daily logging volume.
master_users.csv	id, email, password_hash, salt, full_name, role, zone, assigned_states, is_active, created_at	Credentials database. Passwords secured using PBKDF2-HMAC-SHA256 (100,000 rounds) with per-user 16-byte random salts.
master_issues.csv	id, club_id, club_name, emp_id, reported_at, issue_notes, status, next_reminder_date, resolved_at, resolution_notes	Engine for 72-hour employee reminders, 7-day recurrent cycles, and 30-day Admin escalations.

4.2 Regional Zonal Jurisdiction Roster

User ID	Regional Officer	Zone	Territorial Jurisdiction (36 States & UTs)
EMP01	Rohan Sharma	NORTH	Jammu & Kashmir, Ladakh, Uttarakhand, Himachal Pradesh, Chandigarh, Punjab, Haryana, Delhi, Uttar Pradesh
EMP02	Pooja Verma	CENTRAL	Madhya Pradesh, Chhattisgarh
EMP03	Amit Patel	WEST	Rajasthan, Gujarat, Maharashtra, Goa, Daman and Diu, Dadar & Nagar Haveli
EMP04	Debabrata Ghosh	EAST	Bihar, Jharkhand, West Bengal, Odisha
EMP05	Mayanglambam Singh	NORTH EAST	Sikkim, Assam, Arunachal Pradesh, Meghalaya, Manipur, Tripura, Nagaland, Mizoram
EMP06	K. Venkatesh	SOUTH	Andhra Pradesh, Telangana, Karnataka (South Sector 1)
EMP07	Ananya Nair	SOUTH	Tamil Nadu, Puducherry, Kerala, Andaman & Nicobar Island, Lakshadweep (South Sector 2)

Node Database Structure

Each officer maintains a dedicated local node database directory at `data/employees/empXX/` containing `clubs.csv` (their authorized clubs) and `activity_log.csv` (their daily interaction log). All entries are automatically synchronized to the Central Master databases via the Node-to-Master Sync Engine.

5. Master Login Gateway — Enterprise Secure Access

🔒 Master Login Gateway Overview

The Master Login Gateway is the primary security barrier for the entire NDLI Club Management System. It presents a full-viewport **glassmorphic login screen** when accessing the root URL (/). The gateway requires verification of administrator identity before revealing the central portal selector. Regional Employees bypass this screen by navigating directly to /employee .

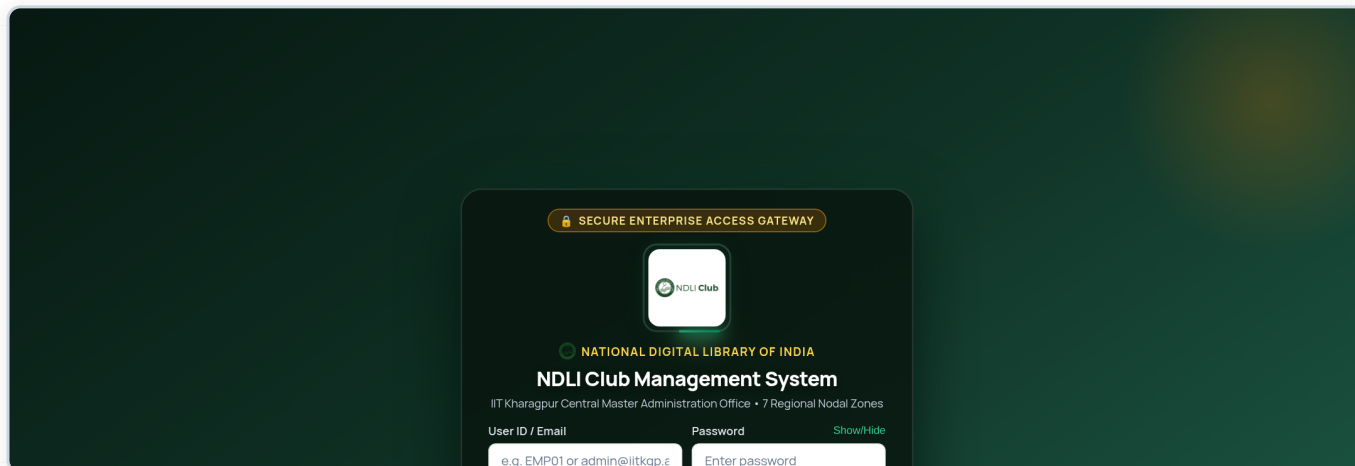


Figure 5.1: Master Login Gateway — Full-Viewport Glassmorphic Security Screen with NDLI Orbit Rings, Live IST Clock & CAPTCHA

5.1 Security Architecture & CAPTCHA Protection

The Gateway employs a layered visual and cryptographic security architecture:

- **PBKDF2-HMAC-SHA256 Password Hashing:** Master Admin passwords are encrypted using 100,000 hashing rounds with per-user 16-byte random salts.
- **Dynamic Canvas Anti-Bot CAPTCHA:** A 5-character randomized alphanumeric challenge is rendered on an HTML5 canvas with random distortion angles, lines, and background noise. It regenerates on each reload or refresh click.
- **Cryptographic Session Tokens:** Upon authentication, a secure 24-byte random session token is stored in `sessionStorage` , automatically expiring when the browser tab is closed.
- **Synchronous Auth Pre-Unlock:** Legitimate ongoing sessions bypass the lock screen instantly via a synchronous script guard, preventing any jerky screen flash during portal navigation.
- **Automated Brute-Force Lockout:** 8 failed sign-in attempts for the same identifier inside 10 minutes trigger an automatic 5-minute lockout (HTTP 429, "Too many failed sign-in attempts"), throttling credential-guessing attacks.
- **Transport & Upload Hardening:** the server-to-server sync/backup transport is relay-key authenticated, certificate signature uploads are validated (image magic bytes, 2 MB cap), and file downloads carry short-lived bearer tokens with automatic stale-token cleanup.

5.2 Step-by-Step: Unlocking the Master System Access

- 1 **Access Root URL:** Open your browser and navigate to `https://ndli-club.onrender.com/` . The glassmorphic lock card appears centered over the forest-green backdrop.
- 2 **Enter Master Identifier:** Input your registered administrator email address: `admin@iitkgp.ac.in` .
- 3 **Enter Master Password:** Type the Master Admin password. Click the **Show/Hide** toggle if needed to visually verify your input.
- 4 **Solve Anti-Bot CAPTCHA:** Read the 5-character code in the security canvas. Type the code (case-insensitive) in the input box. Click ↻ **Refresh** if characters are difficult to read.
- 5 **Unlock Access:** Click the amber gradient "🔒 **Unlock System Access** →" button. The lock screen smoothly fades away in 0.18s, revealing the full Landing Portal.

6. Regional Employee Portal Operations

The **NDLI Club Employee Portal** is the operational workspace for the 7 Regional Zonal Officers. It features real-time clock synchronization, pending reminder feeds, and structured data entry workflows.

6.1 Employee Login, Quick-Select Chips & Zero-Scroll Ergonomics

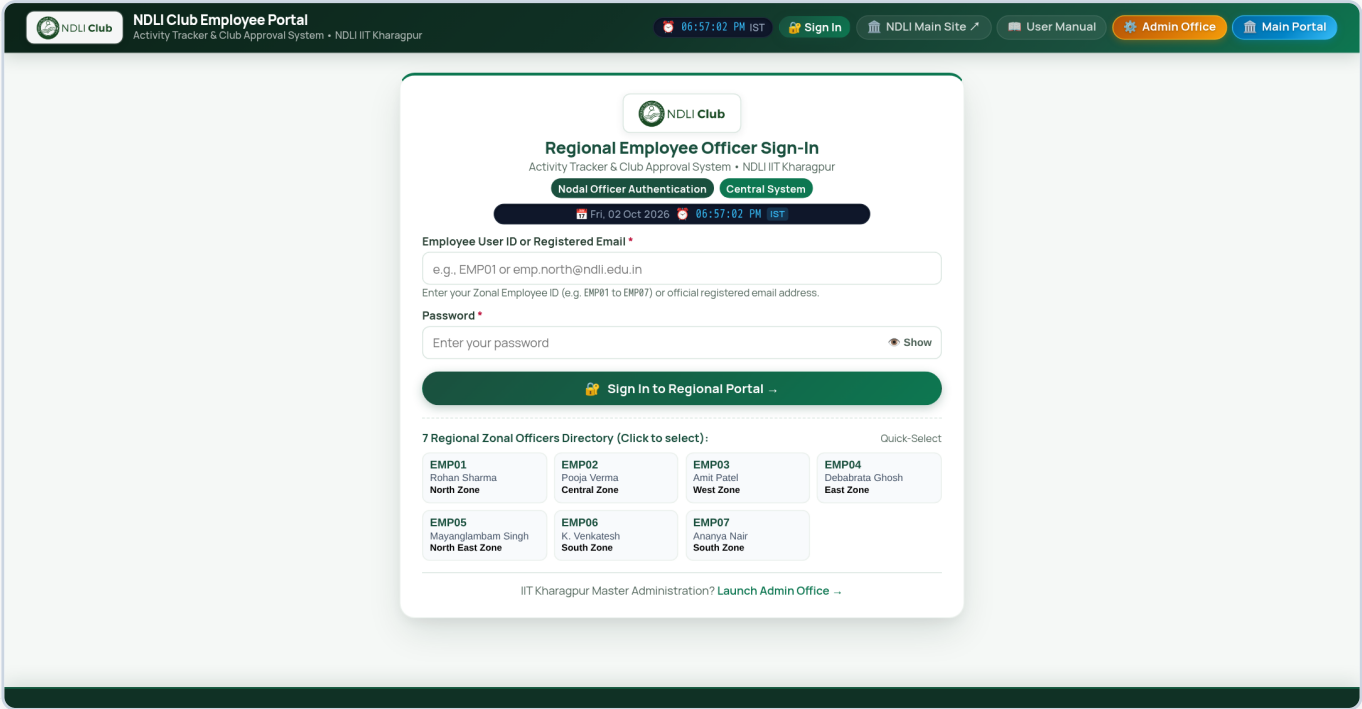


Figure 6.1: Regional Employee Sign-In — Interactive Quick-Select Officer Chips Directory with Live IST Clock

- 1
- Navigate to Portal:** Open `/employee` or click **"Sign In to Portal"** on any EMP card in the main landing page.
- 2
- Quick-Select Regional Officer Chip:** Click any of the 7 officer chips (e.g., *EMP01 — Rohan Sharma — North Zone*). The User ID is automatically filled into the login form.
- 3
- Password Authentication:** Enter the regional password issued to you by your administrator and click **"Sign In to Regional Portal →"**.
- 4
- Workspace Activation:** An encrypted 24-byte session token is generated, loading the officer's dedicated workspace with their node database pre-cached.

Zero-Scroll Ergonomics & Direct Node Shortcuts

The login interface features equalized officer chip grids, compact field spacing, and zero vertical scrolling on standard displays. Officers can also navigate directly via node shortcuts: `/employee/EMP01` through `/employee/EMP07`.

6.2 Step-by-Step: Logging Daily Support Activities (SEC A)

Zonal officers engage with educational institutions daily to resolve obstacles and conduct orientation sessions. Every interaction must be logged immediately in **SEC A — Support Log**:

- 1 **Open Daily Activity Tab:** Confirm you are on the **"Daily Activity"** tab of the Employee Portal.
- 2 **Select Nature of Support:** Choose the appropriate activity tier:
 - **Phone call and remote assistance:** Voice calls, WhatsApp consultations, and remote desktop setup.
 - **Closing of OS Ticket:** Formal resolution of institutional support tickets logged via NDLI helpdesk.
 - **Online training:** Conducting virtual orientation webinars, faculty workshops, and student onboarding.
 - **Offline training:** On-premise physical campus visits, seminars, and library workshops.
- 3 **Activity Notes:** Enter brief notes identifying the institution, key stakeholders, and topics addressed.
- 4 **Submit Entry:** Click **"Submit Activity Log"**. The entry is recorded in `activity_log.csv` with automatic timestamping and synced incrementally to the Central Master Database.

6.3 Step-by-Step: Approving & Onboarding New NDLI Clubs (SEC B & C)

When an institution completes its registration requirements, authorize the club through **SEC B & C**:

- 1 **Trigger Approval Form:** Click the button "**⚡ SEC B: Approve New NDLI Club**" to reveal SEC C with a smooth spring animation.
- 2 **Input Institutional Codes:** Enter unique Club ID (format: `CLUB- [STATE] - [YEAR] - [NUM]`), official Registration Number, and Institution Name.
- 3 **State Selection (Auto-Zone Engine):** Select the institution's state from the dropdown. The system automatically determines and locks the statutory Zone (North, Central, West, East, North East, South), preventing territorial misclassification.
- 4 **Stakeholder Contact Matrix:** Provide verified email addresses for Patron (Head of Institution), President (Faculty Lead), and Secretary (Student Representative).
- 5 **Authorize Club:** Click "**✓ Confirm & Authorize New NDLI Club**". The club is committed to the node database, increments the officer's quota, and syncs to `master_clubs.csv`.

⚠ Mandatory Next Renewal Date Invariant (Rule 10)

Upon club approval, the system enforces **Rule 10**: the upcoming renewal date is automatically set to *exactly 1 calendar year (+1 year)* from the **previous renewal due date** (keeping the club's anniversary from the Date of Approval). If a club is renewed late/overdue, the due date is rolled forward in 1-year steps until it lies in the future. Under no circumstances can a renewal date precede DOA.

6.4 Universal Search, 0ms In-Memory Cache & 50-Row Pagination

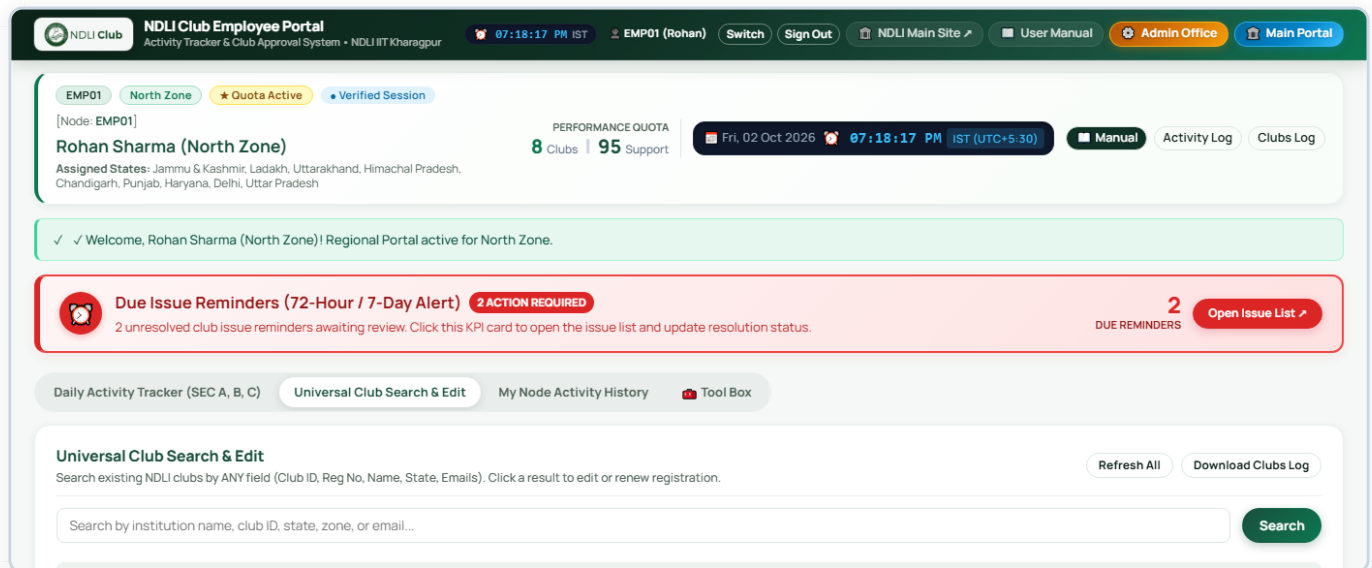


Figure 6.2: Employee Workspace — Universal Search & Edit Engine with In-Memory Caching and 50-Row Pagination

- 1 Instant In-Memory Cache:** The entire club registry is pre-cached client-side (`ALL_CLUBS_CACHE`) upon login. Switching to Tab 2 renders in **0ms** without network latency.
- 2 Sub-35ms Search & 50-Row Pagination:** Typing in the query box filters records instantly in under 35ms. Smooth pagination controls (First, Prev, Next, Last) preserve fluid UI frame rates even with thousands of clubs.
- 3 Inspect, Edit & Annual Renewal:** Click "**Inspect / Edit**" to modify stakeholder emails, or click "**Process Renewal**" to advance renewal by +1 year (from the previous due date) and export vector certificates.

6.5 Node Activity Logs (Tab 3) Auto-Loading & 72-Hour Reminders

Instant Activity Rendering: Navigating to *My Node Activity History* automatically pre-renders from `ALL_ACTIVITIES_CACHE` with 0ms delay (no manual refresh needed). An animated loader appears during live updates, with an inline **Retry** button for connection resilience.

- 72-Hour Grace & 7-Day Reminders:** Reported club issues display active alert banners 72 hours post-logging, recurring weekly until marked "**Resolved**".
- Node CSV Exports:** Officers can click "**Download Activity Log**" and "**Download Clubs Log**" to export instantaneous local CSV archives.

6.6 Integrated Tool Box — 12 Operational Utilities

V4.0 FEATURE

The **Tool Box** tab in the Employee Portal centralizes all daily external services used by Regional Zonal Officers into a single-click launcher. Tools are strategically ordered: operational helpdesk first, followed by Google Workspace services, and concluding with virtual communication platforms.

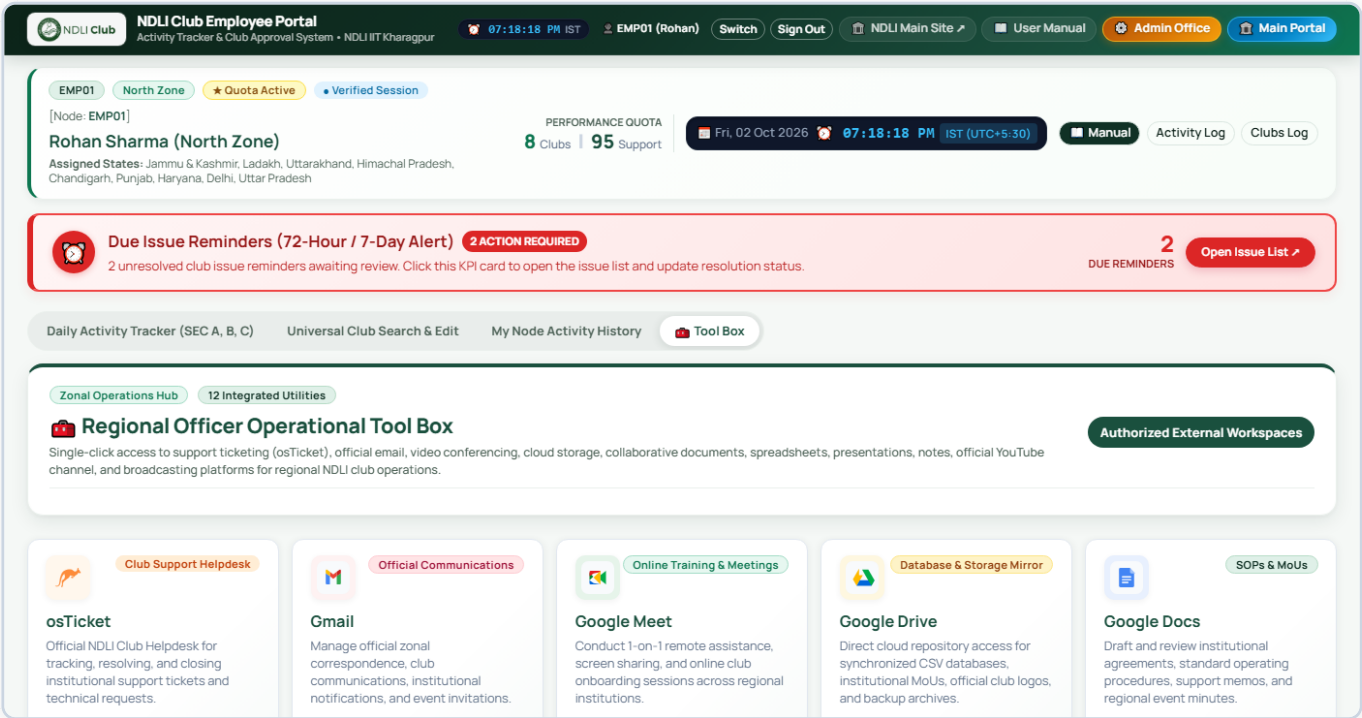


Figure 6.3: Integrated Tool Box Tab — 12 Production Utilities Arranged by Operational Priority (osTicket First, Google Suite, Broadcast)

Complete Tool Directory & Functional Purpose

PRIORITY HELPDESK osTicket Official NDLI Club Helpdesk (club-support.ndl.gov.in). Positioned first for swift issue ticket resolution.	GOOGLE WORKSPACE Gmail Official institutional correspondence, stakeholder notifications, and certificate dispatches.	GOOGLE WORKSPACE Google Meet High-definition virtual meetings for online institutional orientation webinars and training.
GOOGLE WORKSPACE Google Drive Cloud repository for regional club files, presentation decks, and collateral templates.	GOOGLE WORKSPACE Google Docs Collaborative drafting of institutional MoUs, official letters, and inspection notes.	GOOGLE WORKSPACE Google Sheets Tabular management of state-wise target lists, regional rosters, and institutional contacts.
GOOGLE WORKSPACE Google Slides Presentation decks for club orientation seminars and state-level education workshops.	GOOGLE WORKSPACE Google Keep Rapid task checklists, field visit notes, and follow-up memos during campus travel.	GOOGLE WORKSPACE YouTube NDLI video tutorials, user walkthroughs, and national event recordings repository.
VIRTUAL CONFERENCING Zoom Large-scale webinars, multi-institution digital conclaves, and interactive Q&A sessions.	ENTERPRISE COLLABORATION Microsoft Teams Government stakeholder meetings, inter-ministry committees, and regional coordination.	LIVE BROADCAST StreamYard Multi-stream live broadcasting across social channels for nationwide NDLI Club launches.

6.7 Robu — Your NDLI Club AI Assistant

Both portals include **Robu**, a floating AI assistant (bottom-right button) that opens a chat panel. Robu is named after the resourceful robot from Satyajit Ray's *Professor Shonku* stories — built modestly, worth considerably more.

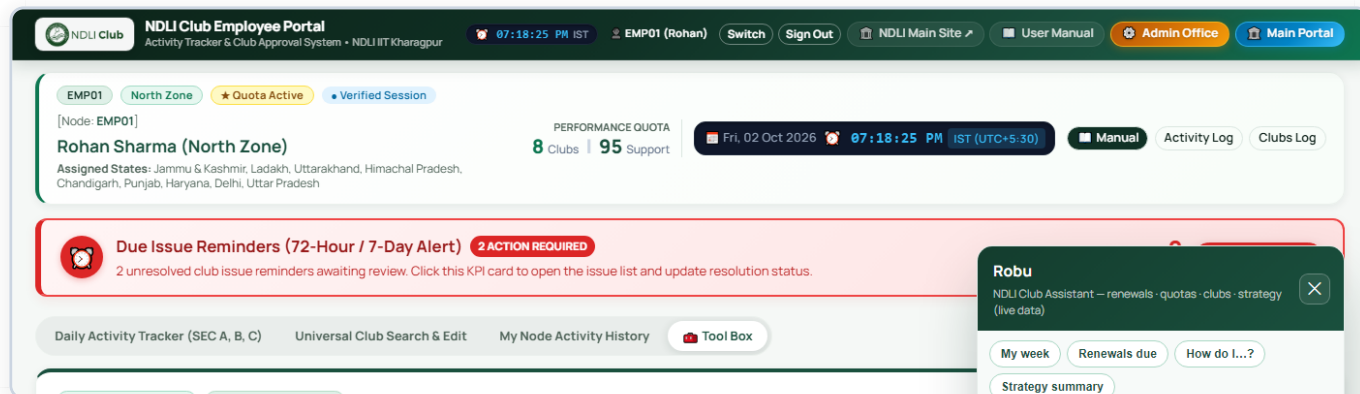


Figure 6.4: Robu in Action — the Floating Launcher Opens the Chat Panel (shown answering "who are you?" with her Professor Shonku origin story)

- **Live data questions:** "How many clubs did I onboard this month?", "Which renewals need attention?" — answered from your own portal data. Regional officers see only their own records; administrators see the national picture. Robu is strictly read-only: she never creates, edits, or deletes anything.
- **Documentation questions:** "How do I approve a renewal?" — answered from this manual and the Technical Reference Manual, with numbered citations so you can verify the source section.
- **Honest limits:** when a question is outside her data or documentation, Robu says so plainly instead of guessing. If no AI language-model key is configured on the server, she operates in clearly-labelled template mode (deterministic answers from the live data).
- **Feedback:** every answer has 👍/👎 buttons — your feedback is recorded to improve future answers.
- **Privacy:** Robu only sees the data your own login is allowed to see; questions and feedback are logged for quality purposes with the platform's standard retention.

7. Central Administration Office Portal

The **Central Administration Office Portal** at IIT Kharagpur is the executive master control console. It synthesizes field reports from all 7 regional zones into actionable statistical intelligence.

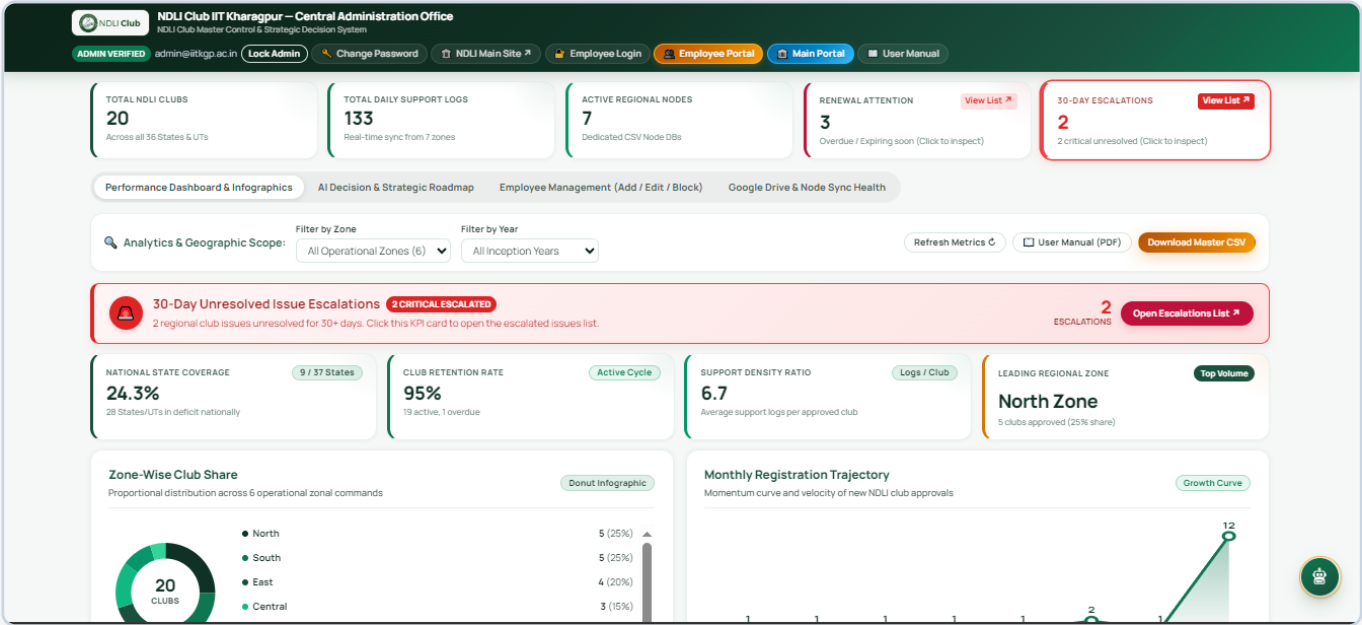


Figure 7.1: Central Administration Executive Dashboard — Real-Time KPIs, Donut Charts & Scope Filters

7.1 Admin Security & Executive KPI Monitors

Administrative access is protected by role-based authentication (`admin@iitkgp.ac.in`). The dashboard displays five top-level operational meters:

- **Total NDLI Clubs:** Cumulative authorized clubs across all 36 States & UTs.
- **Total Daily Support Logs:** Nationwide count of field interactions submitted across all regional nodes.
- **Active Regional Nodes:** Number of operational regional databases communicating in real-time.
- **Renewal Attention Center:** Highlights clubs with expired or imminent registration deadlines (within 30 days). Clicking the card opens an interactive drill-down inspection list.
- **30-Day Escalations Monitor:** Displays persistent club issues unresolved for 30 or more days across any zone, requiring central administrative intervention.

7.2 Statistical Infographics & Geographic Scope Filtering

The dashboard renders dynamic visual charts representing national adoption patterns:

- **Zone-Wise Distribution Donut Chart:** Displays institutional concentration across the 6 geopolitical zones (North, Central, West, East, North East, South).
- **Monthly Trajectory Area Chart:** Illustrates onboarding velocity month-over-month, showing seasonal registration trends.
- **Support Classification Breakdown:** Tracks the nationwide proportion of remote phone support, OS ticket resolutions, online webinars, and physical campus trainings.
- **Scope Filter Bar:** Administrators can filter all dashboard metrics simultaneously by **Zone** (e.g., "North Zone Only") and **Academic Year** (e.g., 2026, 2025).

7.3 Handling 30-Day Administrative Escalations

Issues unresolved at the regional level for more than 30 days automatically trigger central escalation:

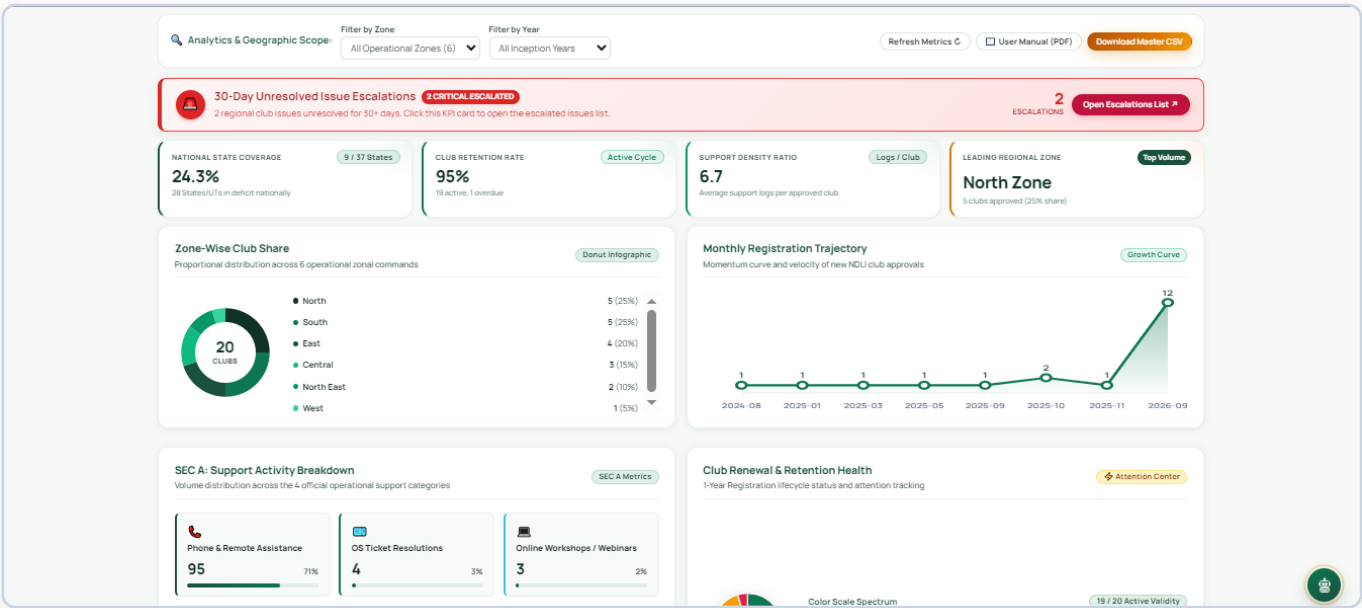


Figure 7.2: Admin Portal Lower Section — Support Activity Breakdown, Renewal & Retention Health, Monthly Registration Trajectory & Zone-Wise Club Share

1. Click the red **"30-Day Escalations"** card in the KPI header.
2. Review the institution's grievance history, originating zonal officer, and days elapsed.
3. Click **"Review & Resolve"** to record Central Office directives and mark the ticket **"Resolved"** or **"Not Resolved"** (re-queuing for 7-day central tracking).

7.4 Employee Performance Evaluation & Star Rating Matrix

The **Employee Performance Evaluation & Star Rating Matrix** provides an objective merit evaluation of all 7 regional zonal officers based on an approved multi-criteria weighting formula:

Composite Score = (Clubs Approved × 0.50) + ((OS Tickets Closed + Phone Calls) × 0.30) + (Offline Training × 0.20) + (Online Training × 0.10)

Evaluation Metric	Weightage	Operational Significance
New Club Approvals	50% WEIGHT	Primary organizational KPI reflecting institutional expansion and onboarding capacity.
OS Tickets Closed & Remote Calls	30% WEIGHT	Measures responsiveness to club operational queries and problem-solving velocity.
Offline Field Training	20% WEIGHT	Physical campus workshops, library orientations, and direct faculty development sessions.
Online Training	10% WEIGHT	Virtual webinars, digital orientation sessions, and remote demonstrations.

Star Rating Thresholds & Categorization

Star Rating	Normalized Benchmark	Performance Appraisal Designation
★★★★★	80% – 100% of Benchmark	Outstanding: Consistently exceeds institutional quotas, fast ticket resolution, proactive field engagement.
★★★★☆	60% – 79% of Benchmark	Commendable: Strong performance across onboarding, ticket closing, and training.
★★★☆☆	40% – 59% of Benchmark	Satisfactory: Fulfills standard institutional quotas and baseline activity targets.
★★☆☆☆	20% – 39% of Benchmark	Needs Improvement: Below expected club onboarding velocity or lagging ticket resolution.
★☆☆☆☆	0% – 19% of Benchmark	Critical Deficit: Immediate administrative consultation and field support required.

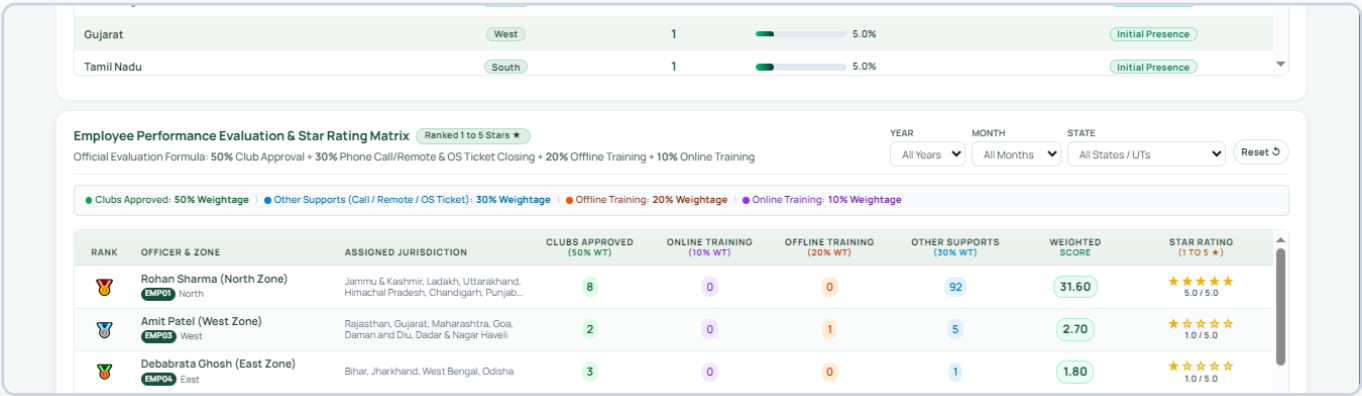


Figure 7.3: Real-Time Employee Star Rating Matrix with Month, Year & State Filters

7.6 AI Strategic Decision Module & Policy Roadmaps

The **AI Strategic Decision Engine** executes continuous heuristic evaluation across the aggregated master databases to generate strategic recommendations for central leadership:

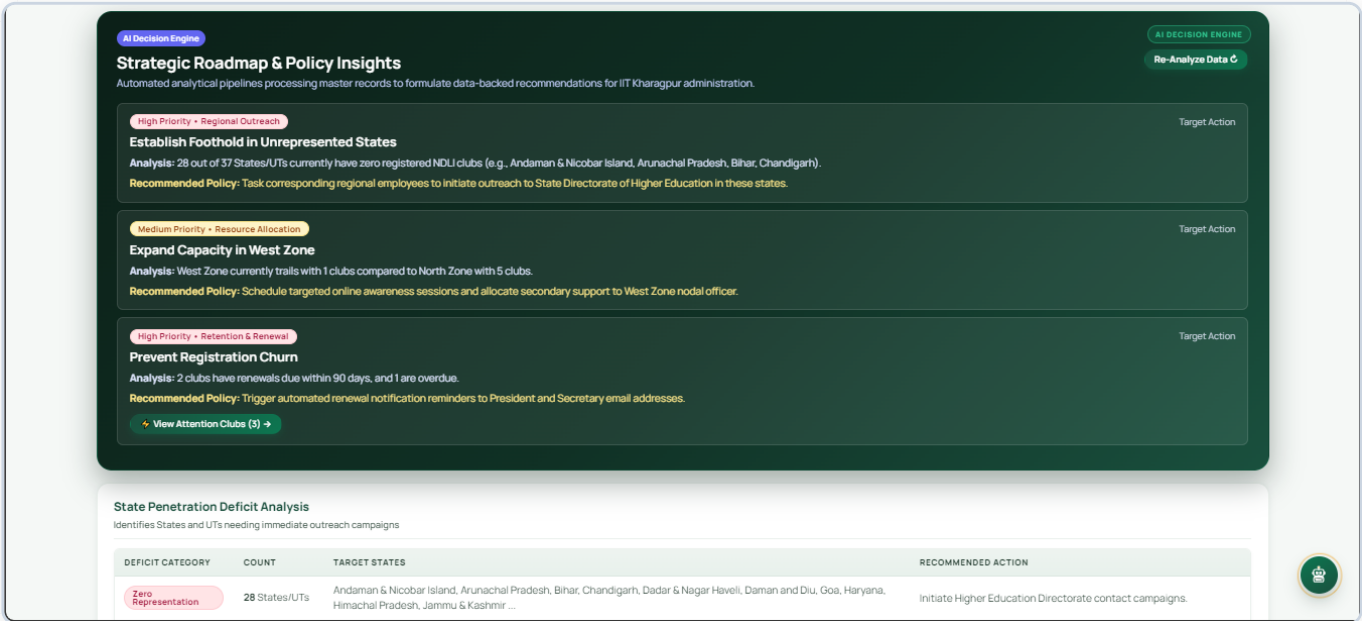


Figure 7.5: AI Strategic Decision Engine — Growth Corridors, Territorial Deficit Formulations & Milestone Roadmaps

- **State Penetration Deficit Formulation:** Automatically identifies territories with club densities below national population-adjusted benchmarks, recommending targeted state-level campaigns.
- **High-Growth Corridor Detection:** Identifies zones exhibiting accelerating adoption momentum to allocate supplemental training materials and regional resources.
- **Milestone Policy Roadmaps:** Synthesizes operational directives into **30-Day Immediate**, **90-Day Mid-Term**, and **1-Year Strategic Milestones**.
- **Automated Resource Allocation Recommendations:** Recommends reallocating training webinars and physical orientation camps to high-potential lagging states based on seasonal academic calendars.

7.7 Node-to-Master Database Reconciliation Engine

The **Master DB Reconciliation** module (operated via the **"Run Full Node-to-Master Reconciliation"** button in the Admin Office Portal) serves as the primary system-wide data integrity and state recovery mechanism for IIT Kharagpur:

- **Distributed Node Audit:** Recursively scans all regional officer node databases across India (`data/employees/emp01/` through `emp07/` , plus any newly provisioned officers), inspecting their local `clubs.csv` and `activity_log.csv` files.
- **Missing Record Discovery & Merge:** Discovers any club approvals or support activities logged during intermittent network connectivity or restored from historical backups, and seamlessly merges them into the Central Master Database without overwriting newer records or losing data.
- **Schema Normalization & Renewal Recalculation:** Validates all date stamps against strict ISO-8601 formatting, verifies stakeholder contact emails, and recalculates dynamic annual club renewal dates.
- **Quota & Star Rating Synchronization:** Recomputes each officer's approved club counts and support activity tallies directly from raw source records, guaranteeing that `master_quotas.csv` and the **Employee Performance Evaluation & Star Rating Matrix** display 100% verified, tamper-proof ratings.

7.8 Automated 7-Day Backups & Google Drive Cloud Mirroring

The platform incorporates an autonomous background backup engine and disaster recovery console:

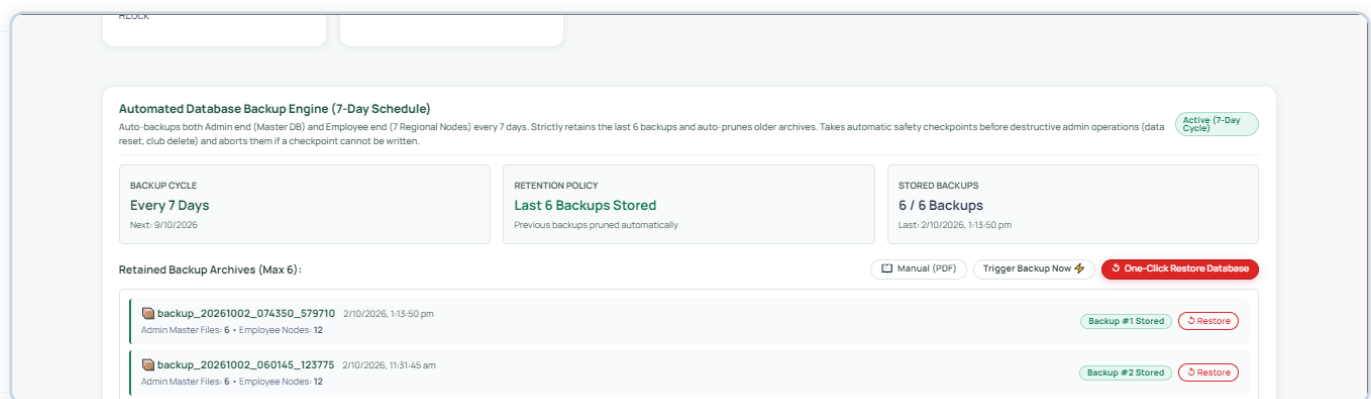


Figure 7.6: Google Drive & Node Sync Health Tab — 7-Day Rolling Backup Status & Cloud Webhook Relay

- **Autonomous 7-Day Cadence:** Archives both Central Master DBs and all 7 Regional Node DBs into compressed ZIP snapshots.
- **Strict 6-Version Retention Invariant:** Preserves only the last 6 historical snapshots. All older archives are automatically pruned, preventing storage bloat.
- **Google Drive Webhook Relay:** Every backup automatically mirrors a timestamped ZIP archive directly into the Google Drive `backups/` repository via Webhook Relay.
- **Signature Files Included:** the authorized signatory signature assets are bundled inside every Drive archive, so a restored backup is immediately certificate-ready.
- **Point-in-Time Drive Copies:** every local backup event — including the automatic checkpoint taken just before a destructive operation (test-data wipe, club deletion, restore) — immediately attempts its own Google Drive copy. The backup console reports the outcome in the `drive_sync` status field ("SUCCESS" only when the Drive relay confirmed receipt; "FAILED" otherwise).
- **Troubleshooting:** relay or sync failures are written to `sync_issues.log` in the data directory together with a loud entry in the server log — an empty log with "SUCCESS" statuses is the healthy state.

7.9 One-Click Emergency Database Restoration & Codebase Scripts

ENTERPRISE

 **Critical Operational Procedure**

The One-Click Emergency Restore operation **completely overwrites both Central Master DBs and all 7 Regional Node DBs** with data from the selected backup archive. Access is strictly gated by a **Dual-Layer Security Protocol**.

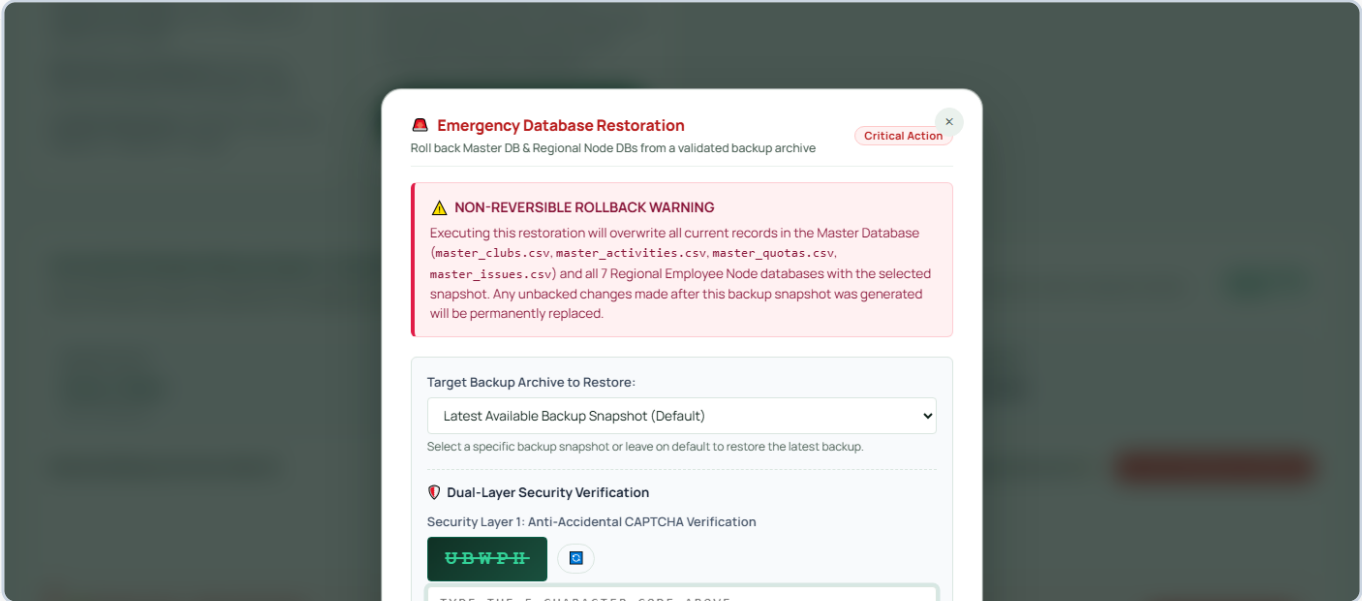


Figure 7.7: Emergency Database Restore Security Modal — Dual-Layer Challenge (Anti-Bot CAPTCHA + Master Admin Password)

The Dual-Layer Security Architecture

LAYER 1

Dynamic Anti-Bot CAPTCHA Challenge

Generates an unforgeable, randomized 5-character alphanumeric token rendered directly in an isolated HTML5 Canvas and validated server-side.

LAYER 2

Master Administrator Password Re-Authentication

Executing database rollback strictly requires re-authenticating as an active administrator: the entered credential is verified against the stored PBKDF2-HMAC-SHA256 hash of an ACTIVE admin account (no fallback or seed credentials are accepted), defeating session hijacking.

 **Codebase-Level One-Click Emergency Restore Utilities**

For code-level emergency recovery, the system provides one-click restore scripts: `restore_to_stable_v2.bat` and `python restore_stable_v2.py` , instantly rolling back the application to the certified Most Stable Version 2 (`stable-v2` , commit `3f864f1`).

7.10 Employee Management & Dynamic Certificate Customization

7.10.1 Regional Employee Account Lifecycle Management

The Central Administration Portal provides centralized control over all Regional Officer accounts:

Admin Authenticated

Verified access to Employee Roster, Node Provisioning, and Access Control.

Lock Section

Add New Employee (Single-Click Provisioning)

Automatically generates new employee credentials, dedicated CSV node database, and regional interface.

Admin Provisioning

Employee ID *

Full Name *

Official Email (User ID) *

e.g. EMP08

e.g. Ramesh Chandra

e.g. emp.south3@ndli.edu.in

Pre-Generated Password *

Regional Zone *

Assigned States *

Min 8 characters

North Zone

e.g. Kerala, Tamil Nadu

Provision Employee & Dedicated Node DB

Employee Credential Database & Access Control

View, edit, block, or unblock employee access. Blocked users cannot log in.

Refresh Roster

USER ID	FULL NAME	EMAIL	ZONE	CLUBS APPROVED	SUPPORT LOGS	STATUS	ACTIONS	INTERFACE
EMP01	Rohan Sharma (North Zone)	emp.north@ndli.edu.in	North	8	95	Active	Edit Block Unblock	Open UI
EMP02	Pooja Verma (Central Zone)	emp.central@ndli.edu.in	Central	3	2	Active	Edit Block Unblock	Open UI
EMP03	Amit Patel (West Zone)	emp.west@ndli.edu.in	West	2	12	Active	Edit Block Unblock	Open UI
EMP04	Debabrata Ghosh (East Zone)	emp.east@ndli.edu.in	East	3	1	Active	Edit Block Unblock	Open UI

Figure 7.8: Central Administration — Regional Employee Roster, Access Toggles & Account Provisioning Form

- **Instant Provisioning:** Enter full name, official email, assigned zone, and covered states to generate a new officer node directory (e.g., emp08), dedicated CSV databases, and credentials.
- **Access Modification & Blocking:** Single-click "Block" button immediately suspends an officer's access without deleting their historical performance quota or support activity logs.
- **Direct Node Impersonation:** Click "Open UI" to launch the employee's portal view in a new tab for quality audits and administrative verification.

7.10.2 Dynamic Renewal Certificate Customization

Administrators can configure the layout and visual assets of official NDLI Club Verification Certificates, including authorized signatory designations, high-resolution PNG signatures, and the official IIT Kharagpur institutional seal:

Renewal Certificate Generation & Download Terminal

High-Resolution (300 DPI A4 Portrait) Official NDLI Club IIT Kharagpur Renewal Certificate

Print-Ready 300 DPI

Club: NDLI-KA-10

Certificate Target Data (From Master CSV)

Live High-DPI A4 Portrait Master Canvas

Institution Name: Indian Institute of Science Bengaluru

Registration No: REG-2024-KA-007

Date of Approval: 01-05-2025

Renewal Validity Upto: 01-05-2027

New PI Signature & Affiliation

Upload New PI Signature (PNG / JPG)

Choose File

No file chosen

* Supports transparent PNG or high-contrast JPG.

NDLI Club

CERTIFICATE OF REGISTRATION

Figure 7.9: Dynamic Renewal Certificate Engine — Institutional Seals, Signatory Vectors & Sutradhar Typography

Employee Self-Service (client requirement, 2026-09-26)

Regional officers may generate renewal certificates themselves from their own portal and may also update the Principal-Investigator signature image used on certificates. Admin-side layout and asset configuration above remains available for standardization.

7.11 Modern Minimalist UI & Complementary Navigation System

ENTERPRISE

The user interface incorporates an ergonomic design system across all three portals (Master Landing, Employee Workspace, and Admin Control Center), ensuring visual hierarchy, high contrast, and fluid motion. Since the Round-5 overhaul, every colour, spacing step, corner radius and elevation level comes from one shared **design-token** set, feedback is delivered through **toast notifications**, and the floating **Robu assistant** launcher (\$6.7) sits in the corner of both working portals:

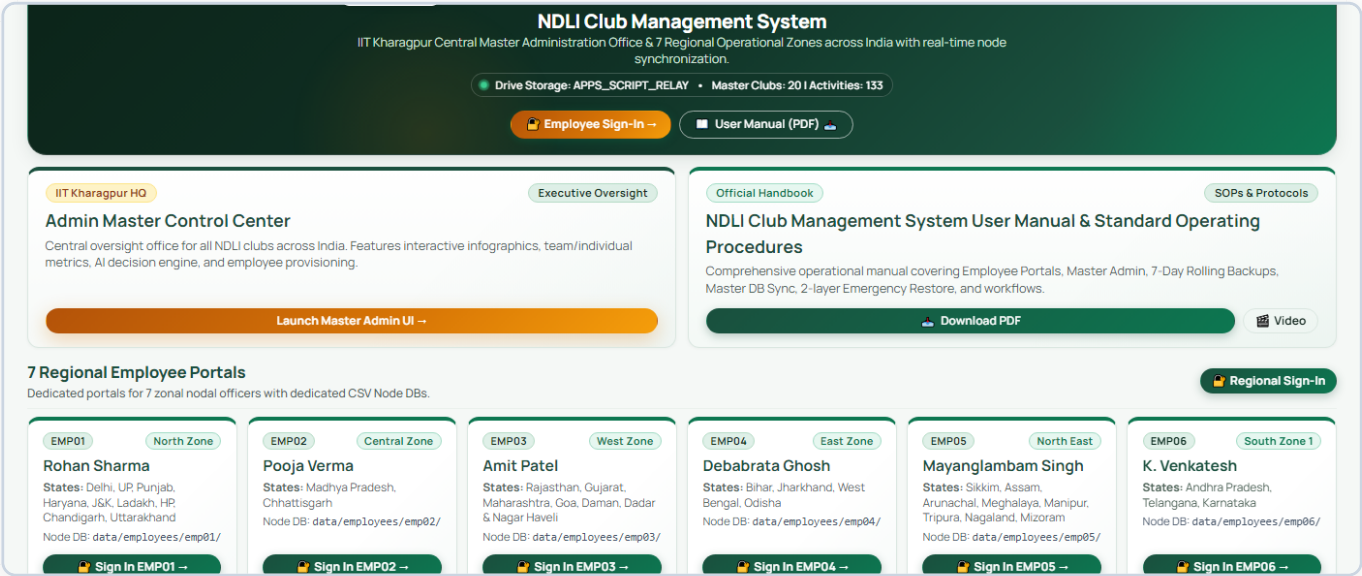


Figure 7.10: Modern Minimalist Landing Gateway — Manrope Typography, Amber Accent CTAs & Staggered Portal Cards

Design System Tokens & Architectural Elements

abc

Manrope Typography

Google Fonts *Manrope* (geometric sans-serif with superior legibility) paired with *Inter* for dense numerical tables and data matrices.

Complementary Palette & Amber CTAs

High-contrast complementary color styling on navbar action buttons ("Employee Login", "Employee Portals") and warm amber CTAs for immediate visual recognition.

Glass Morphism Surfaces

Master lock card and modal overlays utilize deep-frosted glass effects with `backdrop-filter: blur(28px)`, ambient diffusion, and multi-layered borders.

Zero-Scroll Login & Chip Equalization

Compact form heights and equal-width officer chip grids eliminate vertical scrolling on standard viewports, maximizing active workspace visibility.

Motion System: Spring Physics & Scroll-Reveals

Modals open with spring physics (`cubic-bezier(0.34, 1.56, 0.64, 1)`), buttons ripple on click, and cards cascade in via an `IntersectionObserver` hook — one consistent motion language across both portals.

Toasts, Focus States & Accessibility

Save/delete confirmations arrive as non-blocking toast notifications (auto-dismiss, action-aware colours). Every interactive element shows a visible focus ring (`ui-focusable`) for keyboard users, dialogs carry ARIA labels and focus traps, and status changes are announced to screen readers.

8. Maintenance, Enterprise 50k Scaling & FAQs

 **Certified Enterprise Capacity Benchmarks (50,000+ Clubs & 200 Concurrency)**

Primary Key Lookups: 0.64ms avg (O(1) hash index across 50,000 records). **New Inserts:** 13.03ms fast-path O(1) append (zero full-file rewrite). **200 Concurrent Tasks:** Handled in 179.2ms (0 errors). **Crash Shield:** 10-tier Windows lock backoff & global UI error boundary.

8.1 File & Directory Topology

```
ndli_club_management/
├── app.py                # Threaded HTTP Server & REST API (512 slots, 35+ endpoints)
├── auth.py              # PBKDF2 Password Manager & Session Token Engine
├── config.py            # Global Ports, Schemas & Scalability Configuration
├── restore_stable_v2.py  # One-Click Codebase Restore Utility (stable-v2)
├── restore_to_stable_v2.bat # Windows One-Click Codebase Restore Batch
├── db/
│   ├── csv_engine.py    # O(1) Key Hash Indexing, Mutex Locking & 10-Tier Retry
│   ├── sync_engine.py   # Node-to-Master Synchronization & Fast ISO Parser
│   ├── storage_adapter.py # Local & Bounded Drive Sync (_UPLOAD_SEMAPHORE=4)
│   └── backup_engine.py  # 7-Day Scheduler, ZIP Archiver & Retention Safeguard
├── tests/
│   ├── test_scaling_benchmark.py      # 50,000 Club Records Benchmark
│   └── test_scale_50k_and_concurrency.py # 200 Concurrent Requests Stress Test
├── docs/
│   ├── user_manual.html               # This document (HTML source)
│   └── NDLI_Club_Management_User_Manual.pdf # Compiled 21-Page User Manual
└── templates/ & deployment_gas/      # Dual-Deployment 100% Synced Portals
```

8.2 Operational Frequently Asked Questions (FAQs)

Q: How does the system scale to 50,000+ clubs and 200 real-time requests?
The CSV engine builds in-memory hash indices (`key_indices`) for O(1) sub-millisecond lookups, appends new rows directly in O(1) (~13ms) without rewriting multi-megabyte files, and handles 200 concurrent HTTP requests via bounded thread queues in <180ms.

Q: What is the function of the "Run Full Node-to-Master Reconciliation" button?
It audits all regional employee nodes, recovers missing records logged offline, normalizes schemas, and recomputes all quota tallies and star ratings to guarantee 100% data integrity without data loss.

Q: In an emergency, how do I restore the database or codebase?
For database: click **"One-Click Restore Database"** in Admin Portal (requires CAPTCHA + Admin password). For codebase rollback: run `restore_to_stable_v2.bat` or `python restore_stable_v2.py` to return instantly to certified stable-v2.